Design and Engineering of an Optical Tachometer for Spinner with Defined Spokes

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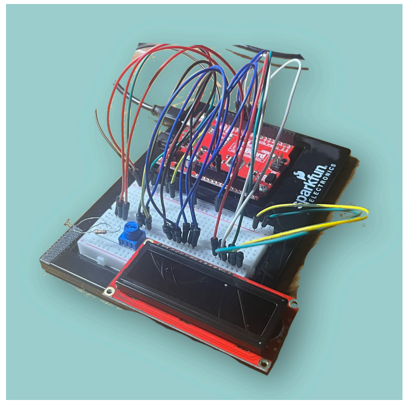
1. **Scope**
    a. Documents the design, being both hardware and software, of a digital tachometer based on an LCD screen, light sensor, external light source, and a spinning object with broad, defined spokes.
    b. A numerical reading for RPM (revolutions per minute) will be output onto the LCD screen every given time interval when a spinner is presented near the light sensor.
    c. The optical tachometer may be used professionally as a non-parasitic gauge for measuring rotational speed (as opposed to a mechanical tachometer, which often robs energy from its spinning component and requires additional hardware). Optical tachometers like this also offer the benefit of use in scenarios involving delicate equipment or dangerous machinery, where mechanical tachometers would not be practical. For example, the tachometer would be helpful in determining the efficiency of ball bearings based on rates of a spinner's rate of RPM change, where minimal interference is critical.
    d. The procedure takes several assumptions including: an opaque spinner (or one with transparency along its rotating surface), a consistent light source, a defined number of broad spokes and gaps (>1mm diameter), and the shadow of the spinner being in line with the light sensor.
2. **Reference Documents**
    a. Statistical References
        i. https://www.ncl.ac.uk/webtemplate/ask-assets/external/maths-resources/statistics/regression-and-correlation/coefficient-of-determination-r-squared.html
            1. (Standard Error Measurements)
3. **Terminology**
    a. Hardware
        i. SparkFun Kit
            1. A breadboard with accompanying sensors, wires, and other devices.
        ii. Tachometer
            1. A device that measures the rotational speed of a spinning object, typically (and in this paper) denoted in RPM.
        iii. Resistor
            1. An object which "resists" the flow of an electric current, causing a drop in voltage.
        iv. LCD Display
            1. Also known as a liquid-crystal display, is a flat panel display used to display characters and data.
        v. Potentiometer

        1. A variable resistor that can modulate the flow of electricity. In this case, it is used to dim or brighten our LCD Display.
    vi. Light Sensor
        1. A form of potentiometer which varies the resistance based on its light input.
  b. *Software*
    i. Arduino
        1. The software (or computer program) controls the tachometer using code.
  c. *Electricity*
    i. Voltage
        1. The force moving an electric current.
    ii. Ohm
        1. A measure of resistance
  d. Data (Formulas are shown in "6. Calculations")
    i. RPM
        1. Revolutions of a spinning object per minute.
    ii. M.D.s.
        1. Manually counted number of times the light in a sensor is dimmed per second.
    iii. M.R.s.
        1. The expected revolutions per second of the spinner based off of the manually counted M.D.s.
    iv. M.R.m.
        1. The expected RPM of the spinner based off of the manually counted M.D.s.
    v. C.R.m.
        1. The measured RPM by the tachometer.
    vi. ± Error
        1. The difference between the measured RPM and expected RPM.

4. Apparatus

  a. Materials used for the completion of the given procedure:
    i. SparkFun Kit
        1. Light Sensor
            a. To send a signal to the computer that the light has been dimmed, and that the spinner has made part of its rotation.
        2. LCD Display
            a. Displays the light reading of the sensor and the RPM of the spinning object.
        3. Potentiometer
            a. Adjusts the brightness of the LCD screen.
        4. 10K Ω Resistor
            a. Accompanies the light sensor to ensure proper function.
        5. Other accompanying components (ex. wires, USB, etc)
            a. Miscellaneous components ensure the functionality of the program.



b. Figure 1: Assembled SparkFun Kit.

ii. Other
   1. Spinner (modified fidget spinner is used in this procedure)
      a. A device that the tachometer will be measuring the spin rate of.

b. Figure 2: Modified 3 spoke fidget spinner.

5. Procedure

a. Follow the instructions given in the “SparkFun Inventor’s Kit Version 4.1a” manual.
   i. Assemble and mount baseplate and breadboard, using the provided screws and sticky tape.
   ii. Install the Arduino software using arduino.cc/downloads.
   iii. Install the Sparkfun drivers using https://learn.sparkfun.com/tutorials.
   iv. Then, consecutively install the components for the following projects:
      1. Circuit 4A: LCD “Hello, World!”
      2. Circuit 1C: Photoresistor
   v. Adjust the location of the wires as needed.
b. Write the code for the program with the Arduino software.
   i. Refer to the code at this repository as a reference: https://github.com/SkirtVrumSkirt/Tachometer/tree/main.
c. Upload the Program to the kit.
   i. Click “verify,” then “upload.”
d. Set up one consistent light source, which casts a light sharp enough that the spinning object will be able to cast a defined shadow onto the light sensor.

i. A flashlight or a desk light will work.

6. **Calculations**
    a. The calculations used to measure the RPM of the spinner were similar in the manual calculations and computer calculations (although the code condensed some computations for simplicity).
    b. In order to ensure the accuracy of the data given by the tachometer, several tests were conducted to cross reference measured and expected speed.
        i. A slow-motion video recording using an iPhone recorded both the tachometer's RPM reading, the time (in seconds) of the recording, and the number of times the spinner passed over the light sensor. These three parameters were used to compare the experimental and expected values for RPM.
        ii. Below is a table of the findings. Included are also the formulas used for the calculations.
    c. *Data*
        i. Error Tests
            1. Low Speed (180-300 RPM)

Manual Detections (M.D.s)	**Revolutions per Second (M.R.s)**	**Revolutions per Minute (M.R.m)**	**Computer Revolutions per Minute (C.R.m)**	**RPM Error**
14	4.67	280	300	+20
13	4.33	260	260	0
13	4.33	260	260	0
12	4.00	240	240	0
11	3.67	220	240	+20
10	3.33	200	200	0
10	3.33	200	200	0
9	3.00	180	180	0

            2. Medium Speed (500-640 RPM)

M.D.s	**M.R.s**	**M.R.m**	**C.R.m**	**RPM Error**
32	10.67	640	640	0
31	10.33	620	620	0

29	9.67	580	580	0
28	9.33	560	540	-20
26	8.67	520	500	-20

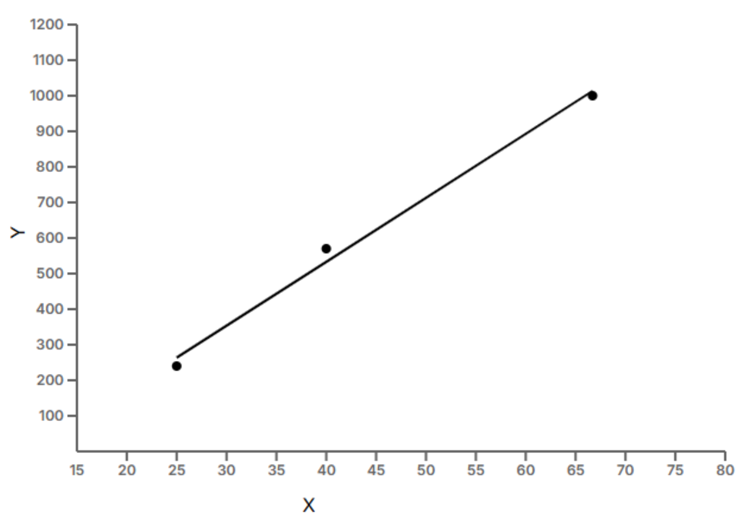
3. High Speed (920-1080 RPM)

M.D.s	**M.R.s**	**M.R.m**	**C.R.m**	**RPM Error**
55	18.33	1100	1080	-20
52	17.33	1040	1040	0
50	16.67	1000	1020	+20
49	16.33	980	980	0
47	15.67	940	960	+20
47	15.67	940	920	-20

4. Error Rates

Test	**Error Rate**
Low Speed	25%
Medium Speed	40%
High Speed	66.7%

5. Figure 3 Linear Regression Curve of Tachometer RPM with Respect to Error Rate.



a. X-axis: Error Rate (%)
    i. Denoted as E.R.
b. Y-axis: AVG RPM for Test
    i. Denoted as A.RPM
c. Slope = 17.98 ± 1.528 (A.RPM/E.R.)
    i. For every 1% increase in the tachometers error rate, the average spin rate increased 17.98 RPM.
d. Y-Intercept = -185 ± 72.06 (E.R.)
e. R^2 (Goodness of Fit): 0.9928 (Very strong)

ii. Calculations & Variables (Some figures were introduced in the terminology section, but will be again for convenience).

1. (*f*), (*i*) = Subscripts for final and initial
2. E.R.E. = Expected Rate of Error (for given test)
3. M.R.E. = Measured Rate of Error (for given test)
4. A.R.E. = Average Rate of Error (for given test)
5. n = Number of tests

6. M.R.s. = (M.D.s) ÷ (spokes)
    a. The expected revolutions per second of the spinner based off of the manually counted M.D.s.

7. M.R.m. = (60 sec/min) x (M.D.s) ÷ (spokes)
    a. The expected RPM of the spinner based off of the manually counted M.D.s.

8. ± Error = (C.R.m) - (M.R.m)
    a. The difference between the measured RPM and expected RPM.

9. A.RPM = (RPM_(max) - RPM_(min)) ÷ 2
    a. The average RPM for a given test.

10. E.R. = ∑(Error) ÷ [(t) x (20 RPM)] x 100
    a. The rate that the tachometer is off of the measured RPM.
    b. E = Sum of measurement errors in RPM for a given test.
    c. t = Measurements conducted for a given test.

11. Slope = [A.RPM_(*f*) - A.RPM_(*i*)] ÷ [E.R._(*f*) - E.R._(*i*)]
    a. Slope of the relationship between RPM of a test and the rate of error of the tachometer to observed RPM.

12. A.R.E. = (1 ÷ n) x ∑(E.R.E)
    a. The average rate of error of our tests. This value will be

different depending on the tachometer readings of different tests that are conducted, so it is only useful for measuring R^2.

13. R^2 (Goodness of Fit) = 1 - [∑(E.R.E - M.R.E.)^2] ÷ [∑(E.R.E - A.R.E.)^2]
    a. How closely the data tracks with what is expected. A value closer to one is very accurate, while a value close to zero is not at all accurate.
    b. In this case, we are measuring the correlation between the rate of error of measurement and the tachometer reading.

d. *Data Analysis*

i. The digital optical tachometer for this procedure measured in increments of 20, as did the manual calculations, due to the nature of the formulas and the way in which data was collected. The preciseness of the RPM is merely a factor of the number of spokes on the spinner, with more spokes resulting in more precise measurements. The increment of “20” comes from 60 (seconds in a minute) divided by 3 (spokes). All calculations and analysis of this paper are conducted assuming the use of a three spoke spinner.

ii. Three tests were conducted, being a low speed (180-300 rpm), medium speed (640-500 rpm), and high speed (1080-920 rpm). Multiple measurements from these tests were taken to ensure statistical accuracy.

iii. It is important to remember that with there only being three speed groups, all measurements and tests comparing these groups should be taken as a reference rather than statistical fact. The findings comparing the results intends only to draw out the flaws of the device, rather than to establish concrete data.

iv. Conveniently, the error was never more than 20 RPM, which is the minimum discrepancy able to be calculated. This means we can conduct binary (true/false) data analysis on our error “rate.” The accuracy of the tests seemed to improve as speeds dropped, however, more testing would be required to verify such discrepancies. The rates of varied data for low, medium, and high speed tests were 25.0%, 40.0%, and 66.7%, respectively. Given that there was no consistent under or over reporting by the computer, it is likely that these discrepancies can be attributed to counting errors from the manual calculations, as with some of the faster tests, it would be unclear whether the fidget spinner had yet completed a full revolution.

v. One may notice how the rates of statistical errors seem to increase relatively linearly with increases in the RPM of the tests. Using a linear regression model, setting a y axis to the average RPM of a test (240, 570, and 1000 rpm for low, medium, and high speed tests, respectively, and the x axis being set to percent error rate (measured above), we obtain the relationship between these figures. This is shown in Fig. 3 Linear Regression Curve of Tachometer RPM with Respect to Error Rate.

vi. When cross referencing the measured and expected data, there is very little variation of the values. In fact, our R-squared value for the linear regression curve is 99.28, which tells us that there is a very strong correlation between these values. This tells us that the discrepancy between the tachometer reading and manual reading may be able to be predicted. However, given that only three different tests were conducted, no definite conclusions can be made about the error rate of the tachometer.

vii. One likely hypothesis for the reason behind the error in the tachometer is actually nothing to do with the tachometer itself, but the method by which the manual RPM calculations are made. With more revolutions of the spinner per second, it would suggest that there are more opportunities for the number of revolutions to be "miscounted" by a video review.

viii. In fact, there is a direct linear correlation between the speed of the spinner and the probability for the RPM to be miscounted, discounting factors such as blurriness of the recording. This is because, as speed increases, there is an increasingly high likelihood for the spinner to be passing over the light sensor as the next second starts.

ix. This phenomenon is similar to that of sticking an object in the path of a fan blade. If the fan is spinning slowly, it is unlikely for the object to be struck. As the fan increases speed, however, so do the chances of the object being struck.

x. This can be conceptually thought of as similar to sticking a hand into a spinning fan for a brief moment. For a slow spinning fan, it is unlikely your hand will be struck, but as the fan speeds up, so do your chances of being hit.

xi. This is an issue, as this opens up the door for vagueness over whether the spinner has made a full revolution or not, leading to a miscount of total revolutions by one third (which is how much all of the errors are off by).

xii. These results suggest that manual counting may introduce an error far greater than that of the tachometer, particularly at higher speeds, potentially making the tachometer the more reliable measurement tool.

7. Conclusion

a. In proximity of a direct light source, the procedure produced a digital optical tachometer that is able to accurately and reliably measure the rotational speed of a spinner within 20 RPM, and up to over 1000 RPM. Throughout this procedure, lessons were learned in the challenges that come with the measurement of real world scenarios, statistical methods, measurements of error, and abilities as well as limitations of technology. This procedure would take approximately two hours to replicate, given the work associated with assembling the Sparkfun kit and the programming of the code, as well as installing the respective drivers. The cost is simply the cost of the Sparkfun kit ($99.95) and a flashlight ($2.00), which would total $101.95. This procedure has many practical and professional applications, which would justify both the cost of time and monetary cost.